HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Spring 2023 Midterm Examination

Date: 1 April 2023 Time: 10:00–12:00

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book exam. You are allowed to bring a one-page double-sided cheat sheet.
- It consists of **13** pages and **8** questions. The last two pages of the exam book can be torn off and used as scratch paper.
- Please write your name, student ID, email, lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all the questions within the space provided on the examination paper. You may also use the back of the pages. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3}+\binom{4}{2}$ instead of 16. Avoid use of nonstandard notation such as ${}_nP_k$ or ${}_nC_k$. Calculators may be used for the exam (but are not necessary).

Problems	1	2	3	4	5	6	7	8	Total
Points	15	12	10	13	13	12	15	10	100
Score									

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1: Propositional Logic [15 pts]

(a) [8 pts] Let

1. r: It rains.
2. f: It is foggy.
3. s: The sailing race will be held.
4. l: The life saving demonstration will go on.
5. t: The trophy will be awarded.

First translate the following hypotheses into propositional logic statements:

1. If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on.
2. If the sailing race is held, then the trophy will be awarded.
3. The trophy was not awarded.

Then use rules of inference to prove the conclusion "it rains". For each step, you need to write from which earlier statement this step is derived from, but you don't have to write the name of the rule used.

(b) [7 pts] Show that the following two propositions are logically equivalent by developing series of logical equivalences.

(i) $(p \wedge q) \rightarrow (p \vee q) \rightarrow (r \wedge q)$,

(ii) $((r \wedge q) \vee (q \wedge \neg r)) \wedge ((r \wedge q) \vee (q \rightarrow r))$

Answer: (a) We are given premises $(\neg r \vee \neg f) \rightarrow (s \wedge l)$, $s \rightarrow t$, and $\neg t$. We want to conclude r.

	Step	**Reason**
1.	$\neg t$	Premise
2.	$s \rightarrow t$	Premise
3.	$\neg s$	Modus tollens using (1) and (2)
4.	$(\neg r \vee \neg f) \rightarrow (s \wedge l)$	Premise
5.	$(\neg(s \wedge l)) \rightarrow \neg(\neg r \vee \neg f)$	Contrapositive of (4)
6.	$(\neg s \vee \neg l) \rightarrow (r \wedge f)$	Simplification using (5)
7.	$\neg s \vee \neg l$	Simplification using (2)
8.	$r \wedge f$	Modus ponens using (6,7)
9.	r	Simplification from (8)

(b)

$$
\begin{aligned}
& (\mathbf{p} \wedge \mathbf{q}) \rightarrow (\mathbf{p} \vee \mathbf{q}) \rightarrow (\mathbf{r} \wedge \mathbf{q}) \\
\equiv\ & \neg(p \wedge q) \vee (p \vee q) \rightarrow (r \wedge q) \\
\equiv\ & (\neg p \vee \neg q) \vee (p \vee q) \rightarrow (r \wedge q) \\
\equiv\ & (\neg p \vee p) \vee (\neg q \vee q) \rightarrow (r \wedge q) \\
\equiv\ & T \vee T \rightarrow (r \wedge q) \\
\equiv\ & \mathbf{T} \rightarrow (\mathbf{r} \wedge \mathbf{q}) \\
\equiv\ & (\mathbf{r} \wedge \mathbf{q})
\end{aligned}
$$

$$
\begin{aligned}
& ((\mathbf{r} \wedge \mathbf{q}) \vee (\mathbf{q} \wedge \neg \mathbf{r})) \wedge ((\mathbf{r} \wedge \mathbf{q}) \vee (\mathbf{q} \rightarrow \mathbf{r})) \\
\equiv\ & (\mathbf{r} \wedge \mathbf{q}) \vee ((\mathbf{q} \wedge \neg \mathbf{r}) \wedge (\mathbf{q} \rightarrow \mathbf{r})) \\
\equiv\ & (r \wedge q) \vee ((q \wedge \neg r) \wedge (\neg q \vee r)) \\
\equiv\ & (r \wedge q) \vee ((q \wedge \neg r) \wedge \neg(q \wedge \neg r)) \\
\equiv\ & (\mathbf{r} \wedge \mathbf{q}) \vee \mathbf{F} \\
\equiv\ & (\mathbf{r} \wedge \mathbf{q})
\end{aligned}
$$

Problem 2: Predicate Logic [12 pts]

Determine the truth value of each of these statements and prove your answer. Let the domain be $\mathbb{R}$.

(a) [3 pts] $\forall x \forall y(x^2 + y^2 \geq 2xy)$

(b) [3 pts] $\exists x \forall y(x \geq y^2)$

Are the following statements equivalent? If yes, give a proof. If not, give a counter example specifying two predicates $P(x)$ and $Q(x)$ and a domain that show the statements are not logically equivalent:

(c) [3 pts] $\neg(\exists x(x > 0 \wedge P(x)))$ and $\forall x(x \leq 0 \vee \neg P(x))$

(d) [3 pts] $\forall x(\neg P(x) \wedge Q(x))$ and $\neg \forall x \exists y(P(x) \vee \neg Q(y))$

Answer: (a) True. $x^2 + y^2 \geq 2xy \leftrightarrow x^2 - 2xy + y^2 \geq 0 \leftrightarrow (x - y)^2 \geq 0$. Since $(x - y)^2 \geq 0$, the statement is true for all x and y.

(b) False. For every x, there exists a y, such as $y = \sqrt{|x| + 1}$, that $x < y^2$.

(c) Equivalent. $\neg(\exists x(x > 0 \wedge p(x))) \equiv \forall x(\neg(x > 0) \vee \neg p(x)) \equiv \forall x(x \leq 0 \vee \neg p(x))$. (By De Morgan's Law)

(d) Not equivalent. Counterexample: $P(x) : x > 0$, $Q(x) : x^2 \geq 0$. L.H.S is false but R.H.S is true.

Problem 3: Sets and Functions [10 pts]

Decide if the following sets are countable or uncountable. Let $\mathbf{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.

1. $\mathbf{N}^2$
2. $P(\mathbf{N})$
3. The set of all functions from $\{0, 1\}$ to $\mathbf{N}$
4. The set of all functions from $\mathbf{N}$ to $\{0, 1\}$
5. The set of all functions from $\mathbf{N}$ to $\mathbf{N}$

Answer: (a) Countable; (b) Uncountable; (c) Countable; (d) Uncountable; (e) Uncountable.

Problem 4: Modular Arithmetic [13 pts]

Find all integer pairs (x, y) such that $2x^2 - 5y^2 = 7$, or prove that such integer pairs do not exist. [Hint: Consider mod 8.]

Answer: First of all, obviously y should be odd, otherwise left hand side will be even. Let $y = 2k+1$. Then $y^2 = 4k(k+1)+1 \equiv 1 \pmod 8$, so $5y^2 \equiv 5 \pmod 8$. Doing mod 8 on both sides of the equation, we have

$$\begin{aligned} 2x^2 - 5 &\equiv 7 \pmod 8, \\ 2x^2 &\equiv 4 \pmod 8, \end{aligned}$$

Then consider the parity of x.

a. If x is even. Then $2x^2 \equiv 0 \pmod 8$, not possible.

b. If x is odd. Then $2x^2 = 2(2k+1)^2 = 8k^2 + 8k + 2 \equiv 2 \pmod 8$, not possible, either.

So such pair (x, y) doesn't exist.

Problem 5 Chinese Remainder Theorem [13 pts]

Solve this system of linear congruences:

$$x \equiv 1 \pmod 3$$
$$4x \equiv 1 \pmod 7$$
$$3x \equiv 1 \pmod{11}$$

Answer:

$$1^{-1} \equiv 1 \pmod 3$$
$$4^{-1} \equiv 2 \pmod 7$$
$$3^{-1} \equiv 4 \pmod{11}$$

Multiply the above inverse on the equation's both sizes respectively:

$$1x \equiv 1 \pmod 3 \to x \equiv 1 \pmod 3$$
$$4x \equiv 1 \pmod 7 \to x \equiv 2 \pmod 7$$
$$3x \equiv 1 \pmod{11} \to x \equiv 4 \pmod{11}$$

Use Chinese Remainder Theorem:

$$M_1 = 77, M_2 = 33, M_3 = 21$$
$$y_1 \equiv 77^{-1} \equiv 2 \pmod 3$$
$$y_2 \equiv 33^{-1} \equiv 3 \pmod 7$$
$$y_3 \equiv 21^{-1} \equiv 10 \pmod{11}$$

We have:

$$x = 1 \cdot 77 \cdot 2 + 2 \cdot 33 \cdot 3 + 4 \cdot 21 \cdot 10 = 1192 \equiv 37 \pmod{231}$$

Problem 6: GCD [12 pts] Calculate $\gcd(2^{100}-1, 2^{65}-1)$.

Answer: In fact, for any a, m, n, we can apply the Euclidean algorithm:

$$\gcd(a^m-1, a^n-1) = \gcd(a^m-a^n, a^n-1) = \gcd(a^n(a^{m-n}-1), a^n-1)$$

Writing $a^m - a^n$ as $a^n(a^{m-n}$, we have

$$\gcd(a^m-1, a^n-1) = \gcd(a^n(a^{m-n}-1), a^n-1)$$

Since $\gcd(a^n, a^n-1) = 1$, we get:

$$\gcd(a^m-1, a^n-1) = \gcd(a^{m-n}-1, a^n-1).$$

We can now use this rule iteratively, in a way similar to the original Euclidean algorithm:

$\gcd(2^{100}-1, 2^{65}-1) = \gcd(2^{65}-1, 2^{35}-1) = \gcd(2^{35}-1, 2^{30}-1) = \gcd(2^{30}-1, 2^5-1) = \gcd(2^{25}-1, 2^5-1) = \cdots = \gcd(2^5-1, 2^5-1) = 2^5-1 = 31.$

Problem 7: RSA [15 pts]

(a) [7 pts] Suppose we implement the RSA cryptosystem by choosing two prime numbers $p = 31$ and $q = 43$. We further choose a number $e = 23$, which is relatively prime to $(p-1)(q-1) = 1260$. What is the value of the secret key d? Show your calculations.

(b) [8 pts] Suppose $m = 2$. Alice encodes her message m using her private key d to compute $x = m^d \pmod{n}$, where d is derived from (a). Show the computational steps.

Answer: (a) Find the inverse of $e = 23$ in Z_{1260}:

$$\begin{aligned} 1260 &= 23 \cdot 54 + 18 \\ 23 &= 18 \cdot 1 + 5 \\ 18 &= 5 \cdot 3 + 3 \\ 5 &= 3 \cdot 1 + 2 \\ 3 &= 2 \cdot 1 + 1 \end{aligned}$$

Hence we get:

$$\begin{aligned} 1 &= 3 - 2 \cdot 1 \\ &= 3 - (5 - 3 \cdot 1) \cdot 1 \\ &= 3 \cdot 2 - 5 \cdot 1 \\ &= (18 - 5 \cdot 3) \cdot 2 - 5 \cdot 1 \\ &= 18 \cdot 2 - 5 \cdot 7 \\ &= 18 \cdot 2 - (23 - 18 \cdot 1) \cdot 7 \\ &= 18 \cdot 9 - 23 \cdot 7 \\ &= (1260 - 23 \cdot 54) \cdot 9 - 23 \cdot 7 \\ &= 1260 \cdot 9 - 23 \cdot 493 \end{aligned}$$

The secret $d = -493 \equiv 767 \pmod{1260}$

(b) $p \cdot q = 1333$

$$
\begin{aligned}
2^{2^0} \bmod 1333 &= 2 \\
2^{2^1} \bmod 1333 &= 2^2 \bmod 1333 = 4 \\
2^{2^2} \bmod 1333 &= 4^2 \bmod 1333 = 16 \\
2^{2^3} \bmod 1333 &= 16^2 \bmod 1333 = 256 \\
2^{2^4} \bmod 1333 &= 256^2 \bmod 1333 = 219 \\
2^{2^5} \bmod 1333 &= 219^2 \bmod 1333 = -27 \\
2^{2^6} \bmod 1333 &= (-27)^2 \bmod 1333 = 729 \\
2^{2^7} \bmod 1333 &= 729^2 \bmod 1333 = 907 \\
2^{2^8} \bmod 1333 &= 907^2 \bmod 1333 = 188 \\
2^{2^9} \bmod 1333 &= 188^2 \bmod 1333 = 686
\end{aligned}
$$

$$767 = 2^9 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$$

$$
\begin{aligned}
2^{767} &\equiv 2^{2^9} \cdot 2^{2^7} \cdot 2^{2^6} \cdot 2^{2^5} \cdot 2^{2^4} \cdot 2^{2^3} \cdot 2^{2^2} \cdot 2^{2^1} \cdot 2^{2^0} \\
&\equiv 686 \cdot 907 \cdot 729 \cdot (-27) \cdot 219 \cdot 256 \cdot 16 \cdot 4 \cdot 2 \\
&\equiv (686 \cdot 907 \cdot 729 \bmod 1333) \cdot (-27 \cdot 219 \cdot 256 \bmod 1333) \\
&\quad \cdot (16 \cdot 4 \cdot 2 \bmod 1333) \\
&\equiv 16 \cdot 560 \cdot 128 \\
&\equiv 500 \pmod{1333}
\end{aligned}
$$

Problem 8: Algorithm [10 pts]

For each pair of expressions (A, B) below, indicate whether A is O, Ω, or Θ of B. Note that zero, one, or more of these relations may hold for a given pair. List all correct ones. It often happens that some students will get the directions wrong, so please write out the relation in full, i.e., $A = O(B)$, $A = \Omega(B)$, or $A = \Theta(B)$ and not just $O(B)$, $\Omega(B)$, or $\Theta(B)$, omitting the A.

(a) [2 pts] $A = n^{2.9} + 100n^2$, $B = n^3$;

(b) [2 pts] $A = 2^{2n}$, $B = 2^{3n}$;

(c) [2 pts] $A = \log n$, $B = \log_5 n$;

(d) [2 pts] $A = 2^{2 \log n}$, $B = n^2$;

(e) [2 pts] $A = \log \log n$, $B = 10^{100}$

Answer: (a) $A = O(B)$

(b) $A = O(B)$

(c) $A = O(B)$, $A = \Theta(B)$, and $\Omega(B)$

(d) $A = O(B)$, $A = \Theta(B)$, and $\Omega(B)$

(e) $A = \Omega(B)$